Lesson 02 material: The Goals of Science; Science and Common Sense

1.4: Goals of Science

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Learning Objectives

1. Describe the three goals of science and give an example for each.
2. Distinguish between basic research and applied research.

The Broader Purposes of Scientific Research in Psychology

People have always been curious about the natural world, including themselves and their behavior (in fact, this is probably why you are studying psychology in the first place). Science grew out of this natural curiosity and has become the best way to achieve detailed and accurate knowledge. Keep in mind that most of the phenomena and theories that fill psychology textbooks are the products of scientific research. In a typical introductory psychology textbook, for example, one can learn about specific cortical areas for language and perception, principles of classical and operant conditioning, biases in reasoning and judgment, and people's surprising tendency to obey those in positions of authority. And scientific research continues because what we know right now only scratches the surface of what we *can* know.

The Three Goals of Science

The first and most basic goal of science is **to describe**. This goal is achieved by making careful observations. As an example, perhaps I am interested in better understanding the medical conditions that medical marijuana patients use marijuana to treat. In this case, I could try to access records at several large medical marijuana licensing centers to see which conditions people are getting licensed to use medical marijuana. Or I could survey a large sample of medical marijuana patients and ask them to report which medical conditions they use marijuana to treat or manage. Indeed, research involving surveys of medical marijuana patients has been conducted and has found that the primary symptom medical marijuana patients use marijuana to treat is pain, followed by anxiety and depression (Sexton, Cuttler, Finnell, & Mischley, 2016).[1].

The second goal of science is **to predict**. Once we have observed with some regularity that two behaviors or events are systematically related to one another we can use that information to predict whether an event or behavior will occur in a certain situation. Once I know that most medical marijuana patients use marijuana to treat pain I can use that information to predict that an individual who uses medical marijuana likely experiences pain. Of course, my predictions will not be 100% accurate but if the relationship between medical marijuana use and pain is strong then my predictions will have greater than chance accuracy.

The third and ultimate goal of science is **to explain**. This goal involves determining the causes of behavior. For example, researchers might try to understand the mechanisms through which marijuana reduces pain. Does marijuana reduce inflammation which in turn reduces pain? Or does marijuana simply reduce the distress associated with pain rather than reducing pain itself? As you can see these questions tap at the underlying mechanisms and causal relationships.

Basic versus Applied Research

Scientific research is often classified as being either basic or applied. **Basic research** in psychology is conducted primarily for the sake of achieving a more detailed and accurate understanding of human behavior, without necessarily trying to address any particular practical problem. The research of Mehl and his colleagues falls into this category. **Applied research** is conducted primarily to address some practical problem. Research on the effects of cell phone use on driving, for example, was prompted by safety concerns and has led to the enactment of laws to limit this practice. Although the distinction between basic and applied research is convenient, it is not always clear-cut. For example, basic research on sex differences in talkativeness could eventually have an effect on how marriage therapy is practiced, and applied research on the effect of cell phone use on driving could produce new insights into basic processes of perception, attention, and action.

References

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1.5: Science and Common Sense

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Learning Objectives

1. Explain the limitations of common sense when it comes to achieving a detailed and accurate understanding of human behavior.
2. Give several examples of common sense or folk psychology that are incorrect.
3. Define skepticism and its role in scientific psychology.

Can We Rely on Common Sense?

Some people wonder whether the scientific approach to psychology is necessary. Can we not reach the same conclusions based on common sense or intuition? Certainly we all have intuitive beliefs about people's behavior, thoughts, and feelings—and these beliefs are collectively referred to as **folk psychology**. Although much of our folk psychology is probably reasonably accurate, it is clear that much of it is not. For example, most people believe that anger can be relieved by "letting it out"—perhaps by punching something or screaming loudly. Scientific research, however, has shown that this approach tends to leave people feeling more angry, not less (Bushman, 2002)[1]. Likewise, most people believe that no one would confess to a crime that they had not committed unless perhaps that person was being physically tortured. But again, extensive empirical research has shown that false confessions are surprisingly common and occur for a variety of reasons (Kassin & Gudjonsson, 2004)[2].

Some Great Myths

In *50 Great Myths of Popular Psychology*, psychologist Scott Lilienfeld and colleagues discuss several widely held commonsense beliefs about human behavior that scientific research has shown to be incorrect (Lilienfeld, Lynn, Ruscio, & Beyerstein, 2010)[3]. Here is a short list:

- "People use only 10% of their brain power."
- "Most people experience a midlife crisis in their 40's or 50's."
- "Students learn best when teaching styles are matched to their learning styles."
- "Low self-esteem is a major cause of psychological problems."
- "Psychiatric admissions and crimes increase during full moons."

How Could We Be So Wrong?

How can so many of our intuitive beliefs about human behavior be so wrong? Notice that this is an empirical question, and it just so happens that psychologists have conducted scientific research on it and identified many contributing factors (Gilovich, 1991)[4]. One is that forming detailed and accurate beliefs requires powers of observation, memory, and analysis to an extent that we do not naturally possess. It would be nearly impossible to count the number of words spoken by the women and men we happen to encounter, estimate the number of words they spoke per day, average these numbers for both groups, and compare them—all in our heads. This is why we tend to rely on mental shortcuts (what psychologists refer to as **heuristics**) in forming and maintaining our beliefs. For example, if a belief is widely shared—especially if it is endorsed by "experts"—and it makes intuitive sense, we tend to assume it is true. This is compounded by the fact that we then tend to focus on cases that confirm our intuitive beliefs and not on cases that disconfirm them. This is called **confirmation bias**. For example, once we begin to believe that women are more talkative than men, we tend to notice and remember talkative women and silent men but ignore or forget silent women and talkative men. We also hold incorrect beliefs in part because it would be nice if they *were* true. For example, many people believe that calorie-reducing diets are an effective long-term treatment for obesity, yet a thorough review of the scientific evidence has shown that they are not (Mann et al., 2007)[5]. People may continue to believe in the effectiveness of dieting in part because it gives them hope for losing weight if they are obese or makes them feel good about their own "self-control" if they are not.

Scientists—especially psychologists—understand that they are just as susceptible as anyone else to intuitive but incorrect beliefs. This is why they cultivate an attitude of **skepticism**. Being skeptical does not mean being cynical or distrustful, nor does it mean questioning every belief or claim one comes across (which would be impossible anyway). Instead, it means pausing to consider alternatives and to search for evidence—especially systematically collected empirical evidence—when there is enough at stake to justify doing so. For example, imagine that you read a magazine article that claims that giving children a weekly allowance is a good way to help them develop financial responsibility. This is an interesting and potentially important claim (especially if you have children of your own). Taking an attitude of skepticism, however, would mean pausing to ask whether it might be instead that receiving an allowance merely teaches children to spend money—perhaps even to be more materialistic. Taking an attitude of skepticism would also mean asking what evidence supports the original claim. Is the author a scientific researcher? Is any scientific evidence cited? If the issue was important enough, it might also mean turning to the research literature to see if anyone else had studied it.

Because there is often not enough evidence to fully evaluate a belief or claim, scientists also cultivate a **tolerance for uncertainty**. They accept that there are many things that they simply do not know. For example, it turns out that there is no scientific evidence that receiving an allowance causes children to be more financially responsible, nor is there any scientific evidence that it causes them to be materialistic. Although this kind of uncertainty can be problematic from a practical perspective—for example, making it difficult to decide what to do when our children ask for an allowance—it is exciting from a scientific perspective. If we do not know the answer to an interesting and empirically testable question, science, and perhaps even you as a researcher, may be able to provide the answer.

References

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